

Generation Interconnection System Impact Study Report

For Queue Project Q22-2153

Rustic 138 kV
February 19, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2153. The Interconnection Customer proposes a 110.4 MW solar plus storage facility interconnecting at the Rustic 138 kV substation in the Maryland area. The request was received on August 11, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 579 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,076,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: CANEY CREEK SUBSTATION – BGE TIMONIUM 138 kV
Contingency: CIPRES – THOMPSON PEAK TAP 138 kV
Loading: 120.8% (Rating: 264 MVA)
DFAX: 9.0%
Pre-project loading: 117.1%
2. Facility: IATAN – CIPRES 138 kV
Contingency: NEOGA – CIPRES 138 kV L
Loading: 112.1% (Rating: 181 MVA)
DFAX: 7.6%
Pre-project loading: 107.5%
3. Facility: RAMOS – NEOGA 138 kV
Contingency: DEMOSS – NEOGA 138 kV
Loading: 111.1% (Rating: 251 MVA)
DFAX: 13.2%
Pre-project loading: 105.3%
4. Facility: DEMOSS – TENNEY 138 kV Cir. 2
Contingency: TENNEY – DEMOSS 138 kV Line
Loading: 105.7% (Rating: 324 MVA)
DFAX: 13.0%
Pre-project loading: 101.2%
5. Facility: DEMOSS – RAMOS 138 kV L
Contingency: RAMOS – DEMOSS 138 kV
Loading: 104.4% (Rating: 354 MVA)
DFAX: 13.9%
Pre-project loading: 100.1%
6. Facility: NEOGA – DEMOSS SUBSTATION 138 kV
Contingency: NEOGA – RAMOS 138 kV L
Loading: 102.8% (Rating: 294 MVA)

DFAX: 13.8%

Pre-project loading: 97.7%

7. Facility: TENNEY – RUSTIC 138 kV Line

Contingency: RUSTIC – TENNEY 138 kV Circuit 1

Loading: 100.5% (Rating: 344 MVA)

DFAX: 12.6%

Pre-project loading: 96.5%

8. Facility: TENNEY – RUSTIC 138 kV Circuit 1

Contingency: RUSTIC – TENNEY 138 kV Line 2

Loading: 100.4% (Rating: 344 MVA)

DFAX: 12.6%

Pre-project loading: 96.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BGE TIMONIUM – CANEY CREEK 138 kV line	Rebuild line with higher-capacity conductor	\$1,759,000	25% (\$437,000)
IATAN – CIPRES 138 kV	Rebuild line with higher-capacity conductor	\$1,423,000	45% (\$634,000)
NEOGA – RAMOS 138 kV Line	Rebuild line with higher-capacity conductor	\$2,230,000	59% (\$1,313,000)
DEMOSS – TENNEY 138 kV Cir. 2	Reconductor line	\$259,000	76% (\$197,000)
DEMOSS – RAMOS 138 kV L	Reconductor line	\$666,000	77% (\$510,000)
NEOGA – DEMOSS 138 kV	Reconductor line	\$2,011,000	60% (\$1,206,000)
TENNEY – RUSTIC SUBSTATION 138 kV #2	Reconductor line	\$1,534,000	76% (\$1,167,000)
RUSTIC – TENNEY 138 kV #1	Reconductor line	\$804,000	76% (\$612,000)
	Total allocated to project		\$6,076,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.